

RUPANKAR GHATAK

☎ +91-8420363444 — ✉ rupankar.ghatak18@gmail.com — 🔗 LinkedIn — 🐙 GitHub

SUMMARY

Results-driven Data Engineer experienced in building scalable, production-grade data platforms with real-time streaming, lakehouse architectures, and automated data quality controls. Proficient in developing low-latency ingestion pipelines, complex stream processing, dimensional modeling, and containerized Airflow orchestration.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java
- **Data Processing & Analytics:** PySpark, Apache Spark, dbt, AWS Glue, Pandas, NumPy, Hadoop, HDFS, Hive, Sqoop
- **Data Ingestion & Streaming:** Apache Kafka, Azure Event Hubs
- **Lakehouse Architecture:** Databricks, Delta Lake, Delta Live Tables (DLT), Unity Catalog, Databricks SQL Warehouse, Lakeview
- **Databases & Cloud Platforms:** PostgreSQL, MySQL, MongoDB, Microsoft Azure, AWS
- **DevOps & Monitoring:** Docker, GitHub Actions, ELK Stack (Elasticsearch, Logstash, Kibana)

PROJECTS

Azure Event-Driven Uber Ride Analytics Engine

May 2026 – Jun 2026

Technologies: FastAPI, Azure Event Hubs, Spark Structured Streaming, DLT, Delta Lake, Python, SQL, Azure

- Built a FastAPI event simulator streaming real-time ride data to Azure Event Hubs and used Spark Structured Streaming to ingest live and historical records into Delta Lake Bronze tables with sub-second latency.
- Developed a denormalized Silver OBT with a 3-minute event-time watermark, enabling efficient stream processing and joins across five reference datasets.
- Designed Passenger, Driver, and Trip dimensional models incorporating historical tracking to capture entity and location changes over time.

Real-Time Streaming Fraud Detection Platform

Jul 2026 – Aug 2026

Technologies: Confluent Kafka, Databricks, PySpark, DLT, Unity Catalog, PostgreSQL, Auto Loader, Lakeview

- Deployed a real-time fraud detection pipeline ingesting credit-card transactions from Confluent Kafka into Bronze, Silver, and Gold layers using Databricks Unity Catalog.
- Integrated PostgreSQL customer master data and dynamic JSON watchlist streams through Spark streaming joins to trigger alerts for spend-limit breaches and watchlisted cards.
- Implemented windowed transaction analytics and Databricks Lakeview dashboards to monitor fraud KPIs, transaction volume, and high-risk entities.

E-Commerce Analytics & Data Orchestration Platform

Aug 2026 – Sep 2026

Technologies: Databricks Unity Catalog, dbt, Airflow 3, Docker Compose, Celery, Databricks SQL Warehouse

- Constructed a Medallion Lakehouse transforming six core entities through CDC Bronze ingestion, incremental Silver OBTs, and Gold SCD Type 2 models.
- Engineered modular dbt pipelines using incremental models, snapshots, and automated data-quality validations executed on Databricks SQL Warehouses.
- Constructed containerized Airflow 3 orchestration with Docker Compose and Celery, integrating the Databricks Jobs API for dependency-gated execution and SLA checks.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering

2023 – Present

Techno India University, India

Higher Secondary Education

Apr 2021 – Mar 2023

Air Force School Barrackpore

CERTIFICATIONS & PARTICIPATION

- Walmart USA – Advanced Software Engineering Job Simulation
- Deloitte Australia – Data Analytics Job Simulation
- AI-ML Virtual Internship – AICTE / EduSkills (Supported by Google for Developers)